UNSW CPMSoc

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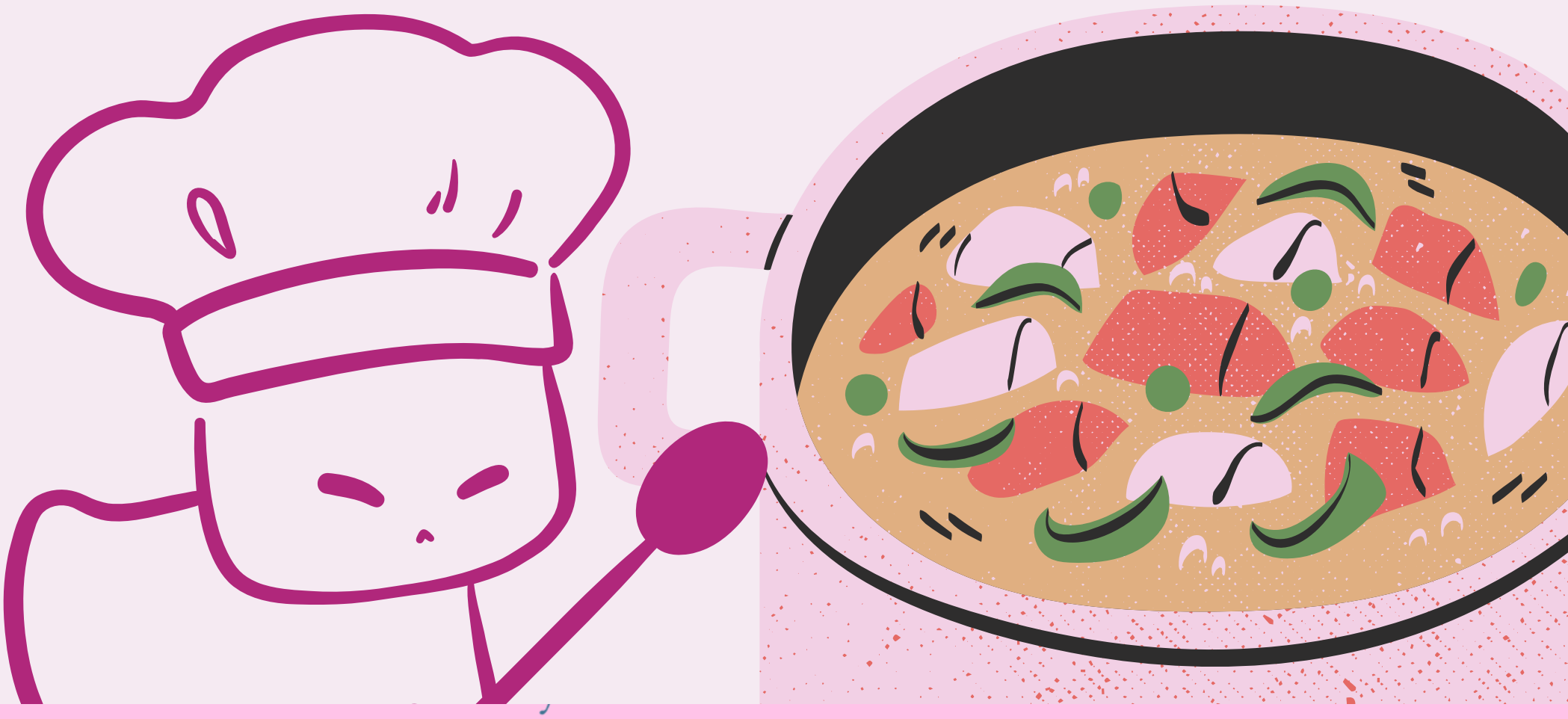
A Contradictory Taste of Competitive Mathematics

Friday, 6th of March 4 - 6pm
Ainsworth G02



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Proof By Contradiciton!

A Collection of Some Delicious Recipes for proof by Contradiction

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Root Vegetable Gratin

SERVINGS: 67 PREPPING TIME: 3 MIN COOKING TIME: 2 MIN

Ingredients

The number $\sqrt{67}$

Prove that this number is irrational

Instructions

1. Assume that $\sqrt{67}$ is rational.
2. Express $\sqrt{67} = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, where a and b have no common divisors.
3. Square the equation and look for a common divisor of a and b, taking extra care with the **prime** 67.
4. Conclude that we obtain a contradiction.

Layer Brownies

SERVINGS: 35 PREPPING TIME: 5 MIN COOKING TIME: 3 MIN

Ingredients

A 6×6 grid of brownies with two opposing corners removed

An unlimited supply of 1×2 sugar icing pieces

Each icing piece covers exactly two brownies; no brownie may be iced twice

The brownies cannot be moved (icing is placed on top)

Instructions

1. Assume that you can cover the brownies.
2. Colour the brownies black and white as on a chessboard.
3. Observe what happens each time we place a 1×2 icing piece.
4. Conclude that this leads to a contradiction.

Pigeon Pie

SERVINGS: m PREPPING TIME: 6 MIN COOKING TIME: 4 MIN

Ingredients

A chef has n holes to store his pigeons

For a real number x, denote $\lceil x \rceil$ as the smallest integer greater than or equal to x. For example:

- $\lceil \frac{15}{4} \rceil = 4$,
- $\lceil 6 \rceil = 6$,
- $\lceil \frac{9}{2} \rceil = 5$

The chef has m pigeons and wants to place each one into a hole

Prove that there is at least one hole containing at least $\lceil \frac{m}{n} \rceil$ pigeons

Instructions

1. Assume that each hole contains at most $\lceil \frac{m}{n} \rceil - 1$ pigeons.
2. Calculate how many pigeons there would be in total under this assumption.
3. Conclude that this leads to a contradiction.

Mousse Cake

SERVINGS: 3 PREPPING TIME: 2 MIN COOKING TIME: 1 MIN

Ingredients

The surface of an infinitely large cake

Every point on the surface must be iced either red or blue according to Antonio's design

Prove that for every distance d one can think of, there exist two points covered with the same colour of icing exactly d apart

Instructions

1. Suppose that this is not true.
2. Consider an equilateral triangle and the colours that its vertices must have.
3. Conclude that this leads to a contradiction.

Carrot Flowers

SERVINGS: ∞ PREPPING TIME: 2 MIN COOKING TIME: 2 MIN

Ingredients

Jeffery has just received his Hōchō knife and is very excited to try it out on his favourite vegetable

His sous-chef challenges him to a duel; the winner keeps the knife

They compete to slice the smallest piece of carrot

Suppose the knife is high-tech enough to slice any positive real amount; prove that there is no smallest possible slice — that is, if left on their own, the chefs will keep slicing without end

Instructions

1. Let $x \in \mathbb{R}^+$ be the minimum amount they can slice.
2. Consider $\frac{x}{2}$; what can we say about this quantity?
3. Conclude that this leads to a contradiction.